

Selected robustness metrics

Continuous values: mean +/- sample SD; control roughness is mean per transition (lower is smoother); roughness/effort/steps: successful trials only

Case / variant	trials	Success	Control roughness overall (mean/transition)	Control roughness after wall (mean/transition)	Control effort	Number of steps	Average time per step (s)
condition=dynamic_wall variant=control_bank_mpp	4	4/4 (100.0%)	0.310 +/- 0.102 (n=4)	0.285 +/- 0.126 (n=4)	278.224 +/- 35.368 (n=4)	58.000 +/- 4.163 (n=4)	0.007 +/- 0.001 (n=4)
condition=dynamic_wall variant=corridor_prior_map	4	3/4 (75.0%)	0.281 +/- 0.014 (n=3)	0.316 +/- 0.013 (n=3)	187.091 +/- 21.710 (n=3)	85.667 +/- 8.737 (n=3)	0.011 +/- 0.003 (n=4)
condition=dynamic_wall variant=frenet_corridor_map	4	4/4 (100.0%)	0.285 +/- 0.037 (n=4)	0.302 +/- 0.039 (n=4)	177.900 +/- 24.570 (n=4)	91.250 +/- 8.139 (n=4)	0.013 +/- 0.003 (n=4)
condition=dynamic_wall variant=full_swarm_prior_map	4	4/4 (100.0%)	0.269 +/- 0.048 (n=4)	0.304 +/- 0.053 (n=4)	176.139 +/- 24.992 (n=4)	97.500 +/- 10.661 (n=4)	0.011 +/- 0.003 (n=4)
condition=dynamic_wall variant=gaussian_prior_map	4	4/4 (100.0%)	0.269 +/- 0.048 (n=4)	0.304 +/- 0.053 (n=4)	176.139 +/- 24.992 (n=4)	97.500 +/- 10.661 (n=4)	0.010 +/- 0.003 (n=4)
condition=dynamic_wall variant=heatmap_prior_map	4	3/4 (75.0%)	0.281 +/- 0.014 (n=3)	0.316 +/- 0.013 (n=3)	187.091 +/- 21.710 (n=3)	85.667 +/- 8.737 (n=3)	0.011 +/- 0.003 (n=4)
condition=dynamic_wall variant=mode_selecting_corridor_map	4	3/4 (75.0%)	0.358 +/- 0.026 (n=3)	0.408 +/- 0.042 (n=3)	250.496 +/- 36.831 (n=3)	67.000 +/- 8.544 (n=3)	0.012 +/- 0.003 (n=4)
condition=dynamic_wall variant=mode_selecting_homotopy_map	4	3/4 (75.0%)	0.358 +/- 0.026 (n=3)	0.408 +/- 0.042 (n=3)	250.496 +/- 36.831 (n=3)	67.000 +/- 8.544 (n=3)	0.012 +/- 0.003 (n=4)
condition=dynamic_wall variant=standard_mpp	4	0/4 (0.0%)	-	-	-	-	0.002 +/- 0.001 (n=4)
condition=no_wall variant=control_bank_map	1	1/1 (100.0%)	0.397 +/- - (n=1)	-	229.250 +/- - (n=1)	51.000 +/- - (n=1)	0.005 +/- - (n=1)
condition=no_wall variant=corridor_prior_map	1	1/1 (100.0%)	0.271 +/- - (n=1)	-	164.921 +/- - (n=1)	76.000 +/- - (n=1)	0.003 +/- - (n=1)
condition=no_wall variant=frenet_corridor_map	1	1/1 (100.0%)	0.227 +/- - (n=1)	-	155.226 +/- - (n=1)	81.000 +/- - (n=1)	0.006 +/- - (n=1)
condition=no_wall variant=full_swarm_prior_map	1	1/1 (100.0%)	0.244 +/- - (n=1)	-	146.955 +/- - (n=1)	84.000 +/- - (n=1)	0.013 +/- - (n=1)
condition=no_wall variant=gaussian_prior_map	1	1/1 (100.0%)	0.244 +/- - (n=1)	-	146.955 +/- - (n=1)	84.000 +/- - (n=1)	0.003 +/- - (n=1)
condition=no_wall variant=heatmap_prior_map	1	1/1 (100.0%)	0.271 +/- - (n=1)	-	164.921 +/- - (n=1)	76.000 +/- - (n=1)	0.003 +/- - (n=1)
condition=no_wall variant=mode_selecting_corridor_map	1	1/1 (100.0%)	0.388 +/- - (n=1)	-	208.358 +/- - (n=1)	58.000 +/- - (n=1)	0.003 +/- - (n=1)
condition=no_wall variant=mode_selecting_homotopy_map	1	1/1 (100.0%)	0.388 +/- - (n=1)	-	208.358 +/- - (n=1)	58.000 +/- - (n=1)	0.003 +/- - (n=1)
condition=no_wall variant=standard_mpp	1	0/1 (0.0%)	-	-	-	-	0.001 +/- - (n=1)
condition=static_wall variant=control_bank_map	4	4/4 (100.0%)	0.338 +/- 0.062 (n=4)	0.338 +/- 0.062 (n=4)	239.266 +/- 23.687 (n=4)	54.500 +/- 3.317 (n=4)	0.007 +/- 0.002 (n=4)
condition=static_wall variant=corridor_prior_map	4	4/4 (100.0%)	0.313 +/- 0.043 (n=4)	0.313 +/- 0.043 (n=4)	193.267 +/- 26.768 (n=4)	99.000 +/- 13.784 (n=4)	0.010 +/- 0.004 (n=4)
condition=static_wall variant=frenet_corridor_map	4	3/4 (75.0%)	0.314 +/- 0.008 (n=3)	0.314 +/- 0.008 (n=3)	175.848 +/- 16.455 (n=3)	93.000 +/- 5.568 (n=3)	0.012 +/- 0.005 (n=4)
condition=static_wall variant=full_swarm_prior_map	4	4/4 (100.0%)	0.299 +/- 0.045 (n=4)	0.299 +/- 0.045 (n=4)	186.567 +/- 24.766 (n=4)	103.750 +/- 7.632 (n=4)	0.010 +/- 0.004 (n=4)
condition=static_wall variant=gaussian_prior_map	4	4/4 (100.0%)	0.299 +/- 0.045 (n=4)	0.299 +/- 0.045 (n=4)	186.567 +/- 24.766 (n=4)	103.750 +/- 7.632 (n=4)	0.010 +/- 0.004 (n=4)
condition=static_wall variant=heatmap_prior_map	4	4/4 (100.0%)	0.313 +/- 0.043 (n=4)	0.313 +/- 0.043 (n=4)	193.267 +/- 26.768 (n=4)	99.000 +/- 13.784 (n=4)	0.010 +/- 0.004 (n=4)
condition=static_wall variant=mode_selecting_corridor_map	4	4/4 (100.0%)	0.379 +/- 0.031 (n=4)	0.379 +/- 0.031 (n=4)	201.113 +/- 5.111 (n=4)	70.500 +/- 8.226 (n=4)	0.015 +/- 0.006 (n=4)
condition=static_wall variant=mode_selecting_homotopy_map	4	4/4 (100.0%)	0.379 +/- 0.031 (n=4)	0.379 +/- 0.031 (n=4)	201.113 +/- 5.111 (n=4)	70.500 +/- 8.226 (n=4)	0.015 +/- 0.005 (n=4)
condition=static_wall variant=standard_mpp	4	0/4 (0.0%)	-	-	-	-	0.003 +/- 0.001 (n=4)

Failure reasons

Counts and percentages among failed trials

Case / variant	Failure reasons
condition=dynamic_wall variant=control_bank_mppi	none
condition=dynamic_wall variant=corridor_prior_mppi	not_reaching: 1 (100.0%)
condition=dynamic_wall variant=frenet_corridor_mppi	none
condition=dynamic_wall variant=full_swarm_prior_mppi	none
condition=dynamic_wall variant=gaussian_prior_mppi	none
condition=dynamic_wall variant=heatmap_prior_mppi	not_reaching: 1 (100.0%)
condition=dynamic_wall variant=mode_selecting_corridor_mppi	not_reaching: 1 (100.0%)
condition=dynamic_wall variant=mode_selecting_homotopy_mppi	not_reaching: 1 (100.0%)
condition=dynamic_wall variant=standard_mppi	not_reaching: 4 (100.0%)
condition=no_wall variant=control_bank_mppi	none
condition=no_wall variant=corridor_prior_mppi	none
condition=no_wall variant=frenet_corridor_mppi	none
condition=no_wall variant=full_swarm_prior_mppi	none
condition=no_wall variant=gaussian_prior_mppi	none
condition=no_wall variant=heatmap_prior_mppi	none
condition=no_wall variant=mode_selecting_corridor_mppi	none
condition=no_wall variant=mode_selecting_homotopy_mppi	none
condition=no_wall variant=standard_mppi	not_reaching: 1 (100.0%)
condition=static_wall variant=control_bank_mppi	none
condition=static_wall variant=corridor_prior_mppi	none
condition=static_wall variant=frenet_corridor_mppi	not_reaching: 1 (100.0%)
condition=static_wall variant=full_swarm_prior_mppi	none
condition=static_wall variant=gaussian_prior_mppi	none
condition=static_wall variant=heatmap_prior_mppi	none
condition=static_wall variant=mode_selecting_corridor_mppi	none
condition=static_wall variant=mode_selecting_homotopy_mppi	none
condition=static_wall variant=standard_mppi	not_reaching: 4 (100.0%)